📄 Unit 2 Classwork — Exponent Rules

Exponent Rules · Power/Zero/Negative Exponents

A. Exponent Rules — Product & Quotient

Problem 1

Simplify: $x^4 \cdot x^3$

x^7

Problem 2

Simplify: $y^8 \div y^3$

y^5

Problem 3

Simplify: $(2x^3)(3x^2)$

$6x^5$

Problem 4

Simplify: $15a^6 \div 3a^2$

$5a^4$

Problem 5

Simplify: $a^2 \cdot a^5 \cdot a$

a^8

B. Power, Zero & Negative Exponents

Problem 1

Simplify: $(x^3)^4$

x^{12}

Problem 2

Simplify: $(2x^2)^3$

$8x^6$

Problem 3

Simplify: 7^0

1

Problem 4

Simplify: $(5x^4)^0$

1

Problem 5

Simplify: 3^{-2} (write as a fraction)

$\frac{1}{9}$

Problem 6

Simplify: x^{-4} (write as a fraction)

$\frac{1}{x^4}$

Problem 7

Evaluate: -4^2 vs $(-4)^2$ — what is $(-4)^2$?

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